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Faculty of Engineering
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Pattern Recognition and Neural Networks

Project Document

Machine Fault Recognition



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Contents

1 Objectives	2
2 Project: Machine Listener System.	2
3 Team Formation	4
4 Research team proposal	4
5 Final Deliverables	4
6 Rules and Instructions	6
7 Grading Criteria	6
8 FAQ	7
9 Delivery Instructions	8

1 Objectives

By the end of this project, you should be able to:

1. Analyze the problem, extract features, and choose the most appropriate preprocessing method (if necessary).
2. Assess performance of different machine learning techniques.
3. Design and implement your own ML pipeline.

2 Project: Machine Listener System.

In this project, you are required to implement a **Mahine Fault Recognition System**. Given an audio file, your system is supposed to classify the audio into one of 6 classes (from 0 to 5). Each audio file represents a recoding of a machine in a factory. There are 3 types of machines and two states (normal/abnormal) for each type

Label	Class
0	Machine 1, Normal
1	Machine 1, Abnormal
2	Machine 2, Normal
3	Machine 2, Abnormal
4	Machine 3, Normal
5	Machine 3, Abnormal

2.1 System Specs

Your system should try to handle variations as much as possible. You should implement a complete machine learning pipeline, including (but not limited to):

- Preprocessing Module.
- Feature Extraction/Selection Module.
- Model Selection and Training Module.
- Performance Analysis Module.

Preprocessing module should handle different audio cases like:

- Background noise
- Volume variations
- Silence removal
- Different microphone quality

You will need to research about the topic, read research papers that tackle this application and similar applications, and do a literature survey that can help you identify the best approaches/ techniques that you can start with in order to improve the accuracy of your results.

2.2 Dataset

The dataset can be downloaded from:

https://drive.google.com/drive/folders/1pFPnn7lbpxWVfOmrDEyvHxXp0_6GFB90

You will divide the dataset into:

- **Training set** to train your model.
- **Validation set** to tune the parameters of your model.
- **Test set** to see how your model performs on unseen data.

3 Team Formation

- A team should be formed of 3 to 4 members. **No team should have more than 4 members under any condition.**
- Please register your team number in the provided spreadsheets.
- All team members should attend the final project discussion. **If one team member failed to show up in the project discussion, then they are regarded as zero contributors in this project.**

4 Research team proposal

Optionally, one of the teams can work on an innovative research idea related to the project problem. The team will be directly supervised by the course instructor, and their main goal is **to publish a research paper** on a new contribution related to the project idea. **The team will not be a part of the competition** and their grades will not be affected by the final leaderboard. However, they will be graded based on their work on the research paper. The teams interested in working on the research project **should submit a proposal to the course instructor** explaining their motivation and proposed contribution. The proposals will be reviewed, then the selected teams can start working on their ideas.

5 Final Deliverables

By the end of this project, each team should submit the following:

1. A **PDF report** that includes the following sections fully and clearly described, here is a recommended flow and template:

<https://www.overleaf.com/read/wbsmbqzkytx#c67e8b>

- Abstract
- Introduction
- System Overview: project pipeline, models/algorithms, system decisions
- Experimental setup: Preprocessing, feature extraction, hyperparameters, system metrics...etc.
- Results: findings, analysis, error analysis (what went wrong).
- Conclusion: Summary of system, limitations, and future work directions.
- The report should include also a workload distribution between team members.

2. **Code** as a zipped folder with the format code [team number].zip containing a readme.md and a requirements.txt. **It is highly recommended to provide an executable file beside the source code.** If you do not provide an executable file, **then you must provide a file called infer.py**, this file will take the path of the “data” directory and produce the output. (Check section 9 to know what the “data” directory contains and what the output should be).
3. **Model file** should be included with your code and in the correct path to execute the code. This model should be loaded into your code upon execution and used to do the test predictions, so make sure that your system can handle different conditions. It’s advised to submit the model file as a Pickle file.
4. **Dockerfile** which will allow us to replicate your environment as accurately as possible so as not to affect you model’s performance.
5. **BONUS:** Deploy your model on any cloud service (Hugging Face is recommended).

6 Rules and Instructions

All students must follow the following rules and instructions:

1. All projects are submitted on the drive.
2. Do not print any document or report. All submissions are electronic.
3. There will be a late penalty for the final project submission.
4. Any sign of cheating or plagiarism will not be tolerated. If one team got caught cheating or plagiarizing from another team, both teams will receive a ZERO for the project grade, in addition to a penalty up to 50% of the project grade.
5. Workload should be distributed fairly and equally. Team members who did not contribute effectively in the project will be penalized.
6. All team members should attend the final discussion. A team member who fails to show up will get a ZERO in the discussion grade.

7 Grading Criteria

The grading criteria of this project is divided as follows:

1. **Report and Discussion: (20%):** This point includes the report submission, the quality of the report and the discussion with all team members. It also evaluates how all team members fully understand the details of the project and can elaborate on the examiner questions.
2. **Project Performance (80%):** Both results accuracy (65%) and running time (15%) will be taken into consideration by the given weights. However, it's not a linear formula but the ranking procedure will be based only on these two factors. Take care that your language choice might affect the running time. This project is a competition-based one.

8 FAQ

1. What programming language(s) can I use?

You may use any programming language of your choice!

2. Is there any restriction on the techniques or approaches to use in any phase of the project?

You are free to use any approach or technique you find appropriate for the problem in hand. It's actually your task to find out the best combination of techniques that will yield the best results. However, you are **NOT limited only to classical machine learning methods** such as *Bayesian Classifiers, KNN, Linear/Logistic Regression, Neural Networks, Support Vector Machines, Principal Component Analysis, etc.*

You are **ALLOWED** to use **deep learning** techniques *e.g. Convolutional Neural Networks, Recurrent Neural Networks, etc.* However, you are limited by a maximum of **2B (billion)** parameters (running at once).

3. How to find research papers for this topic?

There are many resources on the web tackling this problem. You can start by creating an account on [EKB \(Egyptian Knowledge Bank\)](#) with your university provided student email address, which will give you wide access to a huge number of research papers and journals. You can also use the Academic search engines such as [Microsoft Academic](#) or [Google Scholar](#)) a lot of articles are free for public.

4. How will our project be tested?

We will test with our own test set (you have not seen before) in order to standardize the way all teams are graded and ranked. However, you should divide your dataset into Training, Validating and Testing set. The test set is identical to the provided dataset but it is not directly extracted from it.

5. How will our project be ranked?

Please refer to the Grading Criteria section for the grading criteria. The teams will be ranked according to some formula comprising the results accuracy as well as the time taken to generate the results, with the larger weight for the results accuracy. For example, Team X had results with accuracy **92.0%**, with running time = **5 seconds**, will be ranked above team with accuracy **80.0%**, with running time = **2 seconds**.

9 Delivery Instructions

9.1 Input Format

The project will be evaluated using our own test set. You will receive a folder named data. Under this directory, you will find N test audio files. These audio files will be named 1.wav, 2.wav, 3.wav....., 100.wav, 101.wav,....etc.

The hierarchy can be visualized as follows (for the first two test cases):

\data

- 1.wav
- 2.wav
-

Make sure that you process the test files in an **increasingly ordered sequence**. You will output the results in the same order as the input. The submitted model file will be run on all test audio files, so you need to make sure that the system can handle different conditions (preprocess the files).

9.2 Output Format

You should generate **TWO** text files:

1. **results.txt**
2. **time.txt**

9.2.1 Results

In the first file **results.txt**, you should output the prediction of the model for every test case of the N test cases in a single line. The prediction of the model is a number from 0 to 5.

The format of this file should be as follows:

```
2
3
0
2
1
5
3
.
4
1
```

This means that: The the sound in the first file is classified as 2 (Machine 2, Normal), whereas in the second file, it is classified as 3 and so on.

Important Notes:

1. Make sure that you output the result for the test cases **in the same order as the input**. This mistake is very common as you might think you are ingesting the test cases in order, however, you find you are reading the test cases in an incorrect way that you are processing file 10 before 01 and so on. You should print the file names on screen for debugging (remove the print afterwards as it slows down the execution).
2. This file should not include anything other than the model predictions as indicated.
3. Do not include the test case number.
4. Do not include any extra information.
5. Do not print all the test cases in one single line.

9.2.2 Time

In the second file **time.txt**, You should output the time taken (in seconds) by every test iteration (including preprocessing) of the N iterations in a single line per iteration.

The format of this file should be as follows:

```
30.00
35.205
32.320
40.150
35.121
38.982
26.233
.
20.654
```

This means that: For the first test file, your code ran in **30.00 seconds**, for the second test file, your code ran in **35.25 seconds** and so on.

Important Notes:

1. For every iteration, place a timer **after** reading the test audio file, and **after** generating the prediction for the test. You should output the difference between the two times (i.e. the time taken for a single iteration to be processed in the complete pipeline without any I/O overhead).
2. The numbers in the above example are just random numbers and have nothing to do with the actual running times. These numbers are **NOT** a reference and they are just for illustration.
3. The running times should be rounded to **three decimal places**.
4. This file should not include anything other than the model running times as indicated.
5. Do not include the test case number.
6. Do not include any extra information.
7. Do not print all the test cases in one single line.
8. Do not print the results in the console. You have to generate the two files results.txt and time.txt